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Supplementary Materials to "*Bayesian Estimation of Agent-Based Model Parameters Using Normalizing Flows*"

Section A: Hardware Specifications

We ran the estimation pipeline on a machine equipped with a 12-CPU 13th Gen Intel (R) Core (TM) i5-13420H (2.10 GHz) processor, without GPU acceleration. The Keras backend used was 'jax', and the system had 16GB of RAM available.

Section B: Estimating the Modified Bounded Confidence Model with a Different $\tilde{\boldsymbol{x}}$

In this section, we present results obtained using the same Deep Set architecture as in the main manuscript to learn a lower-dimensional embedding of the perturbed opinion distributions. Here, however, the summary representation is expanded to 24 real-valued numbers instead of 16. In other words, the pipeline that transforms the opinion distribution into a lower-dimensional vector encoding its key characteristics now produces a final representation $\tilde{\boldsymbol{x}}$ of dimensionality 24 rather than 16.

Figures S1 and S2 yield results identical to Figures 2 and 3 in the main manuscript. The posterior of ϵ_{max} remains well-calibrated, and the three characteristic regimes are still clearly discernible.

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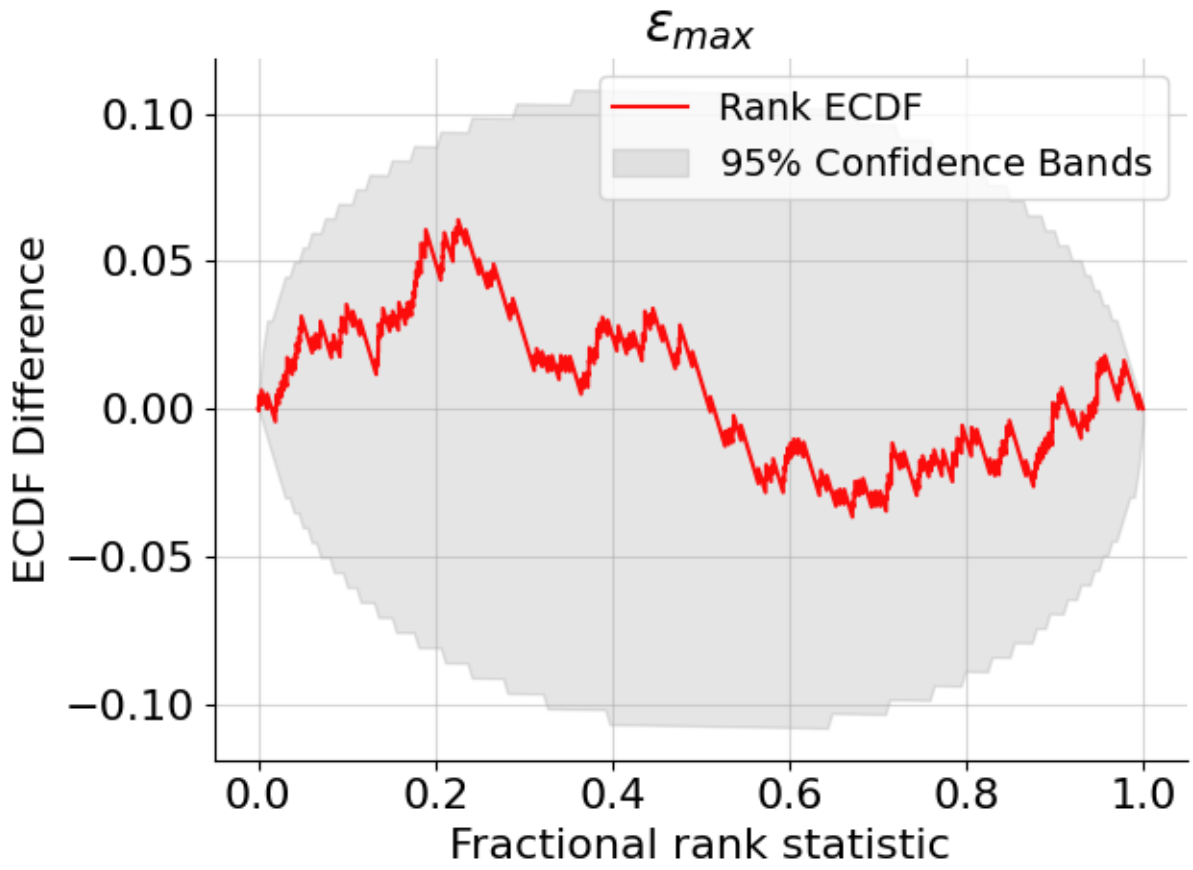


Figure S1: Simulation-based calibration (SBC) results for the posterior distribution of ϵ_{max}, when $\tilde{\boldsymbol{x}}$ is a 24-dimensional vector. The red curve shows the difference between the normalized empirical cumulative distribution function (ECDF) of the rank statistics of the ground truth ϵ_{max} values within their corresponding posterior samples and the cumulative distribution function of a normalized discrete uniform distribution over ranks. The grey bands represent 95% confidence intervals under finite-sample variability. Deviations fluctuating around zero indicate that the inferred posteriors are well calibrated.

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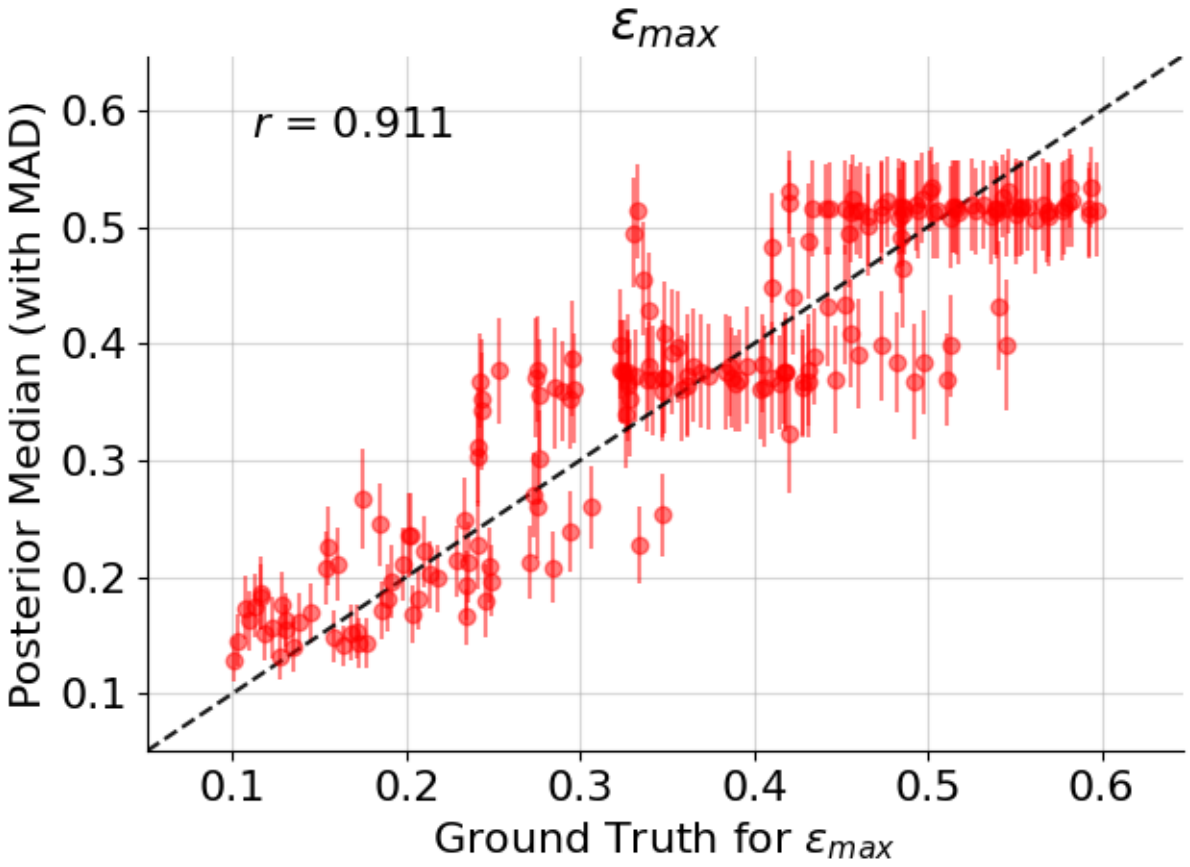


Figure S2: Each point shows a ground truth value of ϵ_{max} plotted against the posterior median inferred from synthetic data generated using that same ground truth value, when $\tilde{\boldsymbol{x}}$ is a 24-dimensional vector. Error bars represent the posterior median absolute deviation (MAD). The dotted diagonal line corresponds to perfect recovery. r denotes the Pearson correlation coefficient between ground truth values and posterior median estimates.

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Section C: Pathological Distributions of Normalized Ranks and Their ECDF Difference Plots - SBC

Figure S3 (top row) shows prototypical pathological distributions of the normalized rank statistics used in Simulation-Based Calibration (SBC) as histograms, while the bottom row displays the corresponding ECDF difference plots. Both the distributions and the ECDF differences indicate the presence of errors in the estimation phase.

Note that these distributions were generated to illustrate specific pathological shapes and were not derived from an actual model. Consequently, no confidence bands are overlaid on the ECDF plots.

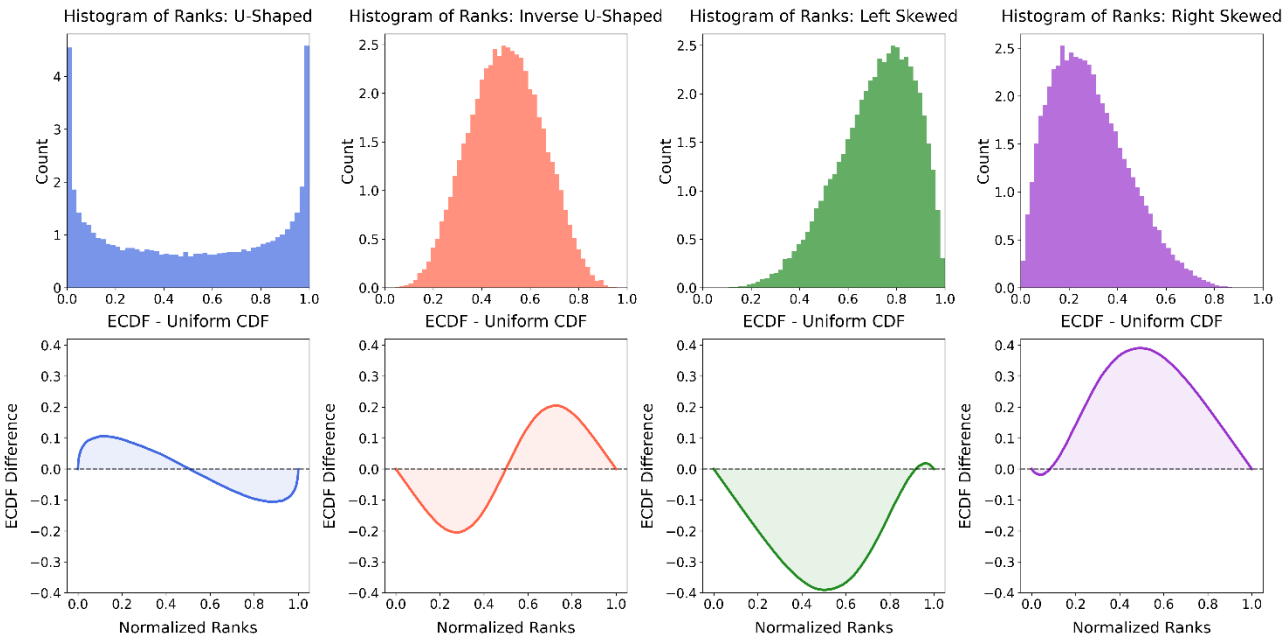


Figure S3: Top row: Histograms of prototypical pathological distributions of the normalized rank statistics used in SBC. Bottom row: Corresponding ECDF difference plots. For simplicity, no confidence bands are overlaid on the ECDF differences.

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Section D: Impact of the Reporting Delay on the SIR Model

To illustrate the differential impact of the reporting delay D on the time series of newly infected cases, we proceeded as follows. We fixed $\lambda = 0.50$, $\mu = 0.10$, and initialized the simulation with 150 "patient zero'" agents. We then generated three scenarios corresponding to $D = \{5, 10, 15\}$. For each scenario, we simulated 500 time series of newly infected cases (Figure S4).

We observe that increasing D causes the observed peak of infections to occur earlier. If reporting of cases is delayed by two weeks while the underlying epidemic dynamics continue to unfold, the observed peak will appear earlier relative to the first report of a case compared to a scenario with $D = 5$. Importantly, the underlying epidemic dynamics remain identical across scenarios. The difference lies solely in the observation window: for $D = 15$, infections are recorded from day 15 to day 65; for $D = 10$, from day 10 to day 60; and for $D = 5$, from day 5 to day 55.

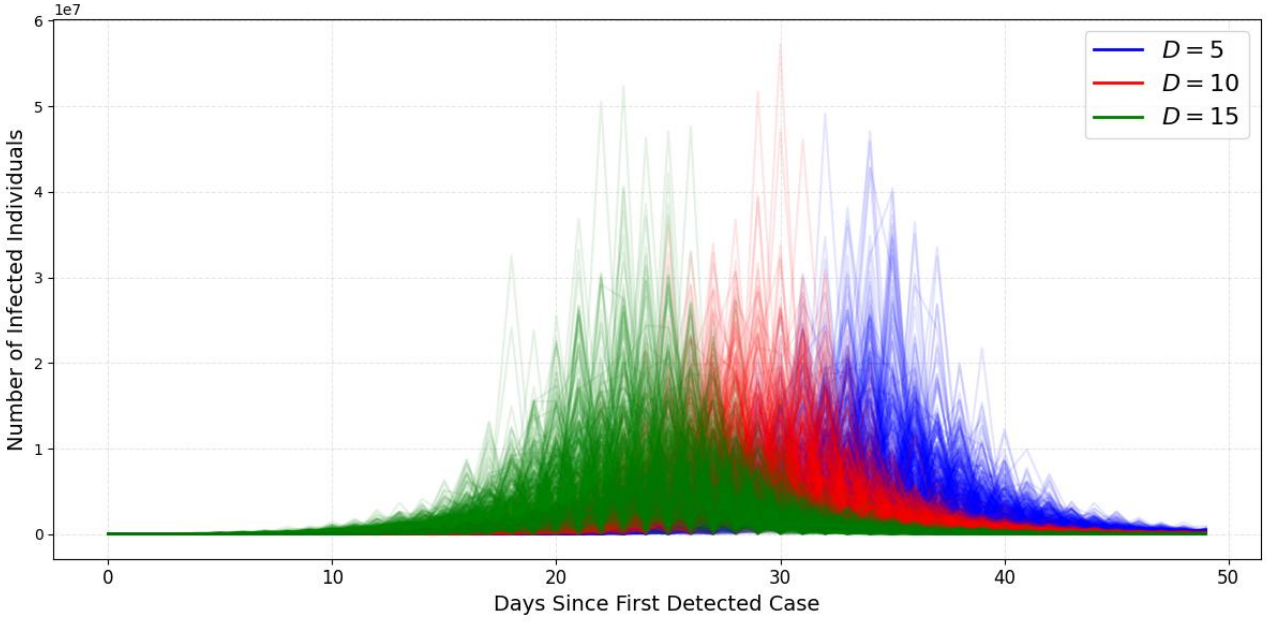


Figure S3: 500 time series of daily new infected cases for different values of D.